

Understanding and Designing for Disaster Preparation on Social Media

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ABSTRACT

The widespread adoption of social media brings both challenges and opportunities for the field of disaster management. However, there has been limited scholarly work on social media usage for disaster preparation. Our work explores how social media sites can be designed to help individuals and communities prepare for disasters, in particular with regard to crowdsourcing information, digital mobilization, and community resilience. We present preliminary findings from 4 online focus groups (N=31), which reveal two emergent themes that may prove useful for future research and design in HCI. We also propose four design recommendations for social media sites as tools for disaster preparation. These findings support the design and evaluation of social media sites to aid in community-based disaster preparedness.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Empirical studies in collaborative and social computing; Collaborative and social computing; Collaborative and social computing theory, concepts and paradigms; Social media; Human computer interaction (HCI); Interaction paradigms; Graphical user interfaces.

KEYWORDS

Crisis informatics, disaster preparation on social media, focus group study, computer-mediated communication, crisis and disaster management, user interface design

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1 INTRODUCTION

Disasters present novel challenges for individuals and communities. The uncertainty and complexity of disasters require individuals to use different platforms and tools to meet their information needs. The ubiquity of new and emerging digital technologies has resulted

in an increased focus on the use of social media platforms to support disaster preparedness in a variety of contexts.

Extant research suggests that social media facilitates peer-to-peer information sharing and enables users to search for timely information from on-the-ground sources in the absence of information from official channels [10, 14, 19]. Social media sites also contribute to the mobilization of community resources during disasters, as individuals turn to their communities for help and support [23]. This suggests that social media can help to strengthen community ties and build resilience during disasters.

While progress has been made in exploring the role of social media in disaster response and recovery, there is limited research on the role of social media in disaster preparedness—in particular, how social media supports communities in using their own resources to prepare for disasters. This work is among the first to explore the design of social media sites to aid in community-based disaster preparedness. Our focus on disaster preparation aims to promote the sustainable development of disaster resilient communities by encouraging community members to develop the skills they need to proactively prepare for and respond to disasters, rather than reacting to events as they occur. We believe that this approach will allow researchers and design practitioners to design and develop tools to help communities implement proactive, rather than reactive, disaster management strategies.

To address this gap, we conducted 4 online focus groups to evaluate how individuals use social media to prepare for disasters, in order to inform the design of tools to support community-based disaster preparedness. We also investigate the use of existing mobile applications for disaster management and derive implications for the design space. Our work surfaces opportunities and challenges for distributed cooperative work during disasters and outlines design recommendations to support community-based disaster preparedness on social media.

2 BACKGROUND AND RELATED WORK

2.1 Background

Community resilience and social capital function as cornerstones of community-based disaster preparedness (CBDP). Community resilience refers to the ability of communities to bounce back from disasters using their own resources [12], while social capital refers to resources in our networks that can be activated through social ties [8]. CBDP encourages at-risk communities to respond quickly and flexibly to natural disasters through local-level, social institutional capacity-building, which in turn equips them with the resources and problem-solving skills they need to withstand future disasters [1, 11].



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Past work surfaces the relationship between social media usage and the formation and maintenance of social capital, such that social media transforms latent ties into weak ties and enables users to seek support or information from their social networks [5, 6]. Our work examines how these affordances can aid in disaster preparedness and the development of disaster-resilient communities. Ellison et al. suggest that Facebook transforms latent ties into weak ties through social context cues, such as mutual friends and shared interests, which can motivate individuals to activate latent ties in their social networks. Activating these latent ties transforms them into weak and bridging ties, which are linked to positive outcomes in terms of bridging social capital.

Vertesi's analytical vocabulary of seams yields additional avenues of potential research surrounding social network sites as heterogeneous, multi-infrastructural environments, such that individuals work with and across different infrastructures to "patch" together or align infrastructural seams [21]. Consequently, the gaps between these heterogeneous environments become seamless or seamful: seamless infrastructures are invisible to users, while seamful infrastructures are visible to users. The multiplicity of social network sites explicates how these infrastructures simultaneously enable and constrain the actions of social media users. Dailey and Starbird extend Vertesi's language of seams to social media use during disasters—in particular, how the alignment of various infrastructures can be at once seamful and seamless [4]. This highlights the multi-infrastructural work of digital volunteers in stitching together social media platforms and audiences to crowdsource information during disasters.

These studies provide a clear rationale for the evaluation of community-based disaster preparedness on social media, such that social media facilitates social capital, which in turn fosters community resilience. We situate our work within a more nuanced understanding of social media sites as discrete multi-infrastructural environments and highlight the emergent need for these platforms to consider different contexts and audiences for disasters.

2.2 Related Work

In this section, we discuss two categories of related work as they pertain to social media usage during disasters: dialogic communication and collective intelligence.

Tang and colleagues distinguish between one- and two-way information sharing on social media during disasters, such that Twitter was more frequently used for one-way information sharing and Facebook for two-way information sharing [20]. Additionally, Facebook was the only social media platform where researchers observed community members commenting on posts from government officials (compared to Twitter and YouTube). Our work builds on this research by exploring how social media platforms can be designed to facilitate two-way information sharing among users during disasters.

Previous research elucidates how disaster management tools might consider the different types of information disseminated on social network sites during disasters while adapting to the specificities of each event [9]. While social media enables users to participate in collective intelligence [3, 7, 15], the ease of access to these sites as information sources explicates the need for a mechanism to filter

through the diversity of available information. This underscores the importance of collective intelligence for disaster management, surfacing the role of online publics in detecting, classifying, and correcting rumors and mitigating the spread of misinformation on social media.

3 METHODS

3.1 Participants

31 participants were recruited from a large university in Southern California using the undergraduate subject pool and offered course credit in exchange for their participation. The participants included 8 men and 23 women, and most were Pre-Communication or Communication majors (71%). 35.5% of participants were juniors, 29.0% were sophomores, 25.8% were freshmen, and the remaining 9.7% were either seniors or graduate students (see Table A.1 for a breakdown of sample characteristics).

3.2 Study Design

4 focus groups were conducted via Zoom. Each workshop was approximately 1.5 hours in length and included 6 to 8 participants, on average. The focus groups were video recorded and transcribed using an automated transcription software called Temi, followed by a manual check to ensure transcription accuracy.

Each focus group consisted of two stages: exploratory research and a design workshop. In the first stage, participants were asked to share their experiences with natural disasters. They were also probed to learn more about how they used social media to prepare for disasters. In the second stage, as part of our design workshop, participants were shown mockups of 3 disaster management mobile applications and asked for feedback on what they liked and disliked and whether they were likely to use the application.

3.3 Materials for Workshop

Screenshots of these applications (Facebook, FEMA, and Emergency by American Red Cross) were edited into device mockups using Figma. These mockups were compiled into slide decks that were shown to participants during each workshop.

3.4 Data Analysis

ATLAS.ti was used to analyze the data. The data was analyzed in three stages: open, axial, and selective coding [18]. First, the interview transcripts were coded line by line to identify raw codes, then they were grouped into categories to indicate relationships among codes, and finally they were collapsed into 2 emergent themes.

4 PRELIMINARY FINDINGS

4.1 The Role of Social Media in Disaster Communication

Our work explicates the role of social media sites as information sources and support networks for communities during disasters.

Social media for information. We found that social media made it easier for participants to work with others in their community to crowdsource information during disasters, which in turn enabled them to maintain situational awareness and make important

decisions (e.g., deciding whether they should evacuate during a wildfire).

P5 shared her experience preparing for a large wildfire:

"Oftentimes if it's a serious fire, my neighbors will all be standing on the street, so we'll all be talking to one another, seeing what each other's plans are to decide what we're going to do" (P5).

While P5 crowdsourced information from her neighbors in person, P31 used social media to replicate the process of collective intelligence online:

"Our area was not under mandatory evacuation, so I still knew people in the area, and I was checking in a lot with them on social media to see how things were going" (P31).

This explicates how the prevalence of social media use during disasters may extend mobilization and collective sensemaking to include digitally networked environments.

Social media for support. Participants also used social media to broadcast requests for information, help, or resources to their social networks during disasters.

P30 expressed her frustration with the lack of information from news outlets during disasters, citing this as the reason why she preferred to get peer-based information directly from friends and family.

"For our situation, the news outlets were probably the least helpful because they got updates very randomly and they would only choose to air them at specific times. It was mainly talking to friends and family and neighbors that gave you the most updated information whenever you needed it" (P30).

This establishes the importance of peer-to-peer information exchange on social networks. Although P30 primarily crowdsourced information from her strong tie connections (e.g., family and friends), past work suggests that the process of collective intelligence may also extend to weak tie connections (e.g., people in the same city).

P24 shared her experience using Snapchat for social support:

"Sometimes people would post about it on their stories. And then I would swipe up and ask them questions, make sure they're okay. Or even myself, if I posted about it, I know other people would often reach out" (P24).

Social media helped participants feel more connected to their communities during disasters, which in turn encouraged them to broadcast information (such as status updates) or requests for help to their social networks. The data from our focus groups indicate that social media sites are unique as tools for disaster management, such that they enable not only distributed work, but also distributed support among social networks. For example, P24 did not have to be in the same location as the connections she interacted with on Snapchat, which highlights the bridging function of social media in mitigating the constraints of geographical proximity for disaster work.

4.2 Distinguishing between Different Social Media Platforms for Disaster Communication

Our research surfaces the role of social media sites as channels for dialogic communication, enabling one- and two-way information sharing among users during disasters. Our findings indicate that social media platforms such as Facebook facilitate two-way information sharing, such that users can exchange information and communicate with others in their social network, whereas platforms such as Twitter facilitate one-way information sharing, such that users can search for information from many different accounts.

P12 preferred to use Twitter for one-way information sharing because it was easier to find real-time information and Instagram for two-way information sharing because she could directly message other users:

"I feel like Twitter I use more for updates and I don't really post anything or communicate on it with anyone. I just look at it. Whereas I feel like on Instagram, I'm communicating with people" (P12).

P27 said that she usually found out about fires by looking at her town's Facebook group to see what other people in her community were saying:

"We got fire warnings. They were sent around the town on our town forum. And then one of our neighbors told us that he would call us, so it was sort of in the news, but more so in the town group on Facebook" (P27).

P3 also felt that Facebook groups were a helpful resource for crowdsourcing information during disasters:

"My city has a Facebook group, so sometimes I'll just go in there and then people will be posting pictures, but I can actually see what it's looking like, which can be helpful" (P3).

Facebook helped our participants mobilize community resources and crowdsource information from social media during disasters. Specifically, Facebook groups enabled participants to aggregate information from people who lived in their town or city, but who they didn't know personally or weren't close with.

Across our 4 groups, participants relied on different social media sites for different information needs, such that they used Facebook for peer-to-peer communication with weak and latent tie connections and Twitter to find up-to-date information from a variety of sources. Participants used Facebook to communicate with latent tie connections (e.g., someone from the same town) with whom they had a prior offline connection, whereas participants used Twitter to search for information from unofficial or informal sources outside of their social networks. Although they perceived these accounts as credible, participants rarely messaged or communicated with them directly.

Overall, our research suggests that the information architecture of social media sites informs their affordances for users by facilitating dialogic communication and mediating how individuals make credibility judgments about information and information sources.

5 DISCUSSION

5.1 Theoretical Implications

Our findings provide support for CBDP approaches to disaster management, as social media facilitates the formation and maintenance of social capital, which in turn supports the development of disaster-resilient communities. Additionally, the present study extends past work on weak- and latent-tie connections to social media usage during disasters [5, 6], such that social media sites enable users to broadcast requests for help or information to everyone in their social networks, regardless of the strength of their social ties. Our work suggests that our reliance on social media as a primary information source during disasters may result in the increasing replication of face-to-face interactions in a digitally networked context, as social media sites allow individuals to reach out to not only their neighbors, but also members of their community who are geographically dispersed.

Additionally, our study extends existing research on peer-to-peer and backchannel communication on social media [10, 19] by examining how communities use social media to prepare for disasters. Participants' preference for social media sites over disaster management applications, such as FEMA and Emergency by American Red Cross, challenges traditional assumptions about authority and expertise in the credibility literature and confirms findings that peer recommendations carry more weight in social media [22]. Ultimately, our study extends new conceptualizations of peer-based credibility as it relates to information sharing during disasters, such that individuals often privilege crowdsourced information from their peers over information from established organizations.

Our findings surface social media sites as infrastructure to support the development and maintenance of social capital during disasters [4, 21]. This conceptualization of social media sites as heterogeneous, multi-infrastructure environments contributes to our understanding of how these platforms support one- and two-way information sharing. This may explain why users look to different platforms for different purposes, e.g., Twitter for real time updates and Facebook for crowdsourcing information from other community members. Our work suggests that social media sites support decision making during disasters in ways that are unique to the specific platform. Because each site focuses on addressing particular user needs, individuals must stitch together multiple social media platforms to meet their diverse information needs during disasters.

While prior work has looked more closely at disaster response and recovery, our study contributes to the HCI literature by highlighting the need for tools to support disaster preparation in at-risk communities, and advances a more holistic understanding of disaster management and distributed cooperative work on social media. Our work also establishes design implications to guide future research and design practice in the field.

5.2 Design Recommendations

Our findings surface four design recommendations to support community-based disaster preparedness on social media.

Weak ties on social media. Our research confirms previous findings in the HCI literature [5, 6], such that the information architecture of social media platforms facilitates connections with weak

ties during disasters, often replacing more direct lines of communication such as text messaging or phone calls. This explicates a need for social media to facilitate the development of social capital among one's weak or latent tie connections, enabling them to access the same resources and information shared with one's strong tie connections. Future research may consider how the design of social media platforms can address more than one user need. For example, Facebook Safety Check may enable users to check in on the well-being of their weak ties, while Facebook Groups may help users crowdsource information across strong and weak tie connections.

Information availability. Our work establishes the importance of information availability as a design consideration for disaster preparation on social media. Because disasters are so time sensitive, ensuring easy access to information is critical to decision making. Due to participants' reliance on their existing social networks, grounded in their local communities, it is more important to leverage the social media tools people use every day with friends and neighbors, rather than relying on specialized or unfamiliar software or platforms where users may lack an existing network.

Visual vs. textual information. Another design consideration is the use of visual versus textual information in user interfaces. Visual information can be processed more quickly than textual information, which is especially useful during disasters when users have to make decisions quickly. Our findings provide support for earlier work on the role of information architecture—specifically, visual information—in digital mobilization during disasters [23]. Visual information such as photos not only supports distributed cooperative work on social media, but also increases the credibility of unofficial information sources. Sites such as Instagram, which are primarily visual, may increase the perceived credibility of users sharing photos and videos of disasters, whereas platforms such as Twitter, which are primarily textual, may increase the perceived credibility of information from credentialed sources (e.g., news outlets). Designing user interfaces that privilege message-level credibility over source-level credibility can encourage the use of social media as a credible information source during disasters.

Reducing information overload. When designing for disaster preparedness, it is important to reduce information overload, as too much information may cause more stress and anxiety for users and make decision making more difficult. This suggests that sites such as Twitter may be more effective information sources during disasters than news outlets, which often publish longer articles with more information. Future research might investigate how the visual hierarchy of user interfaces can prevent information overload.

5.3 Conducting Focus Groups in a Remote Format

Our work also explicates strengths and limitations of focus groups as a method for design research, especially in a remote context. Online focus groups have been established as a viable research method since the late 1990s, and they pose challenges as well as distinctive benefits due to the online environment [16, 17]. Zoom videoconferencing, in particular, has been found to be particularly well-suited for online research, due to its ease of use, low cost, and options for data management and security [2]. The group format allowed participants to work together and brainstorm ideas for new

features. We found that participants often elaborated on ideas that others had suggested or offered alternate perspectives on a topic, which allowed for a range of opinions. The remote environment may have also resulted in an increased willingness to disclose personal information and experiences, as participants were in their homes and able to participate from behind a screen (which may have reduced their anxiety). The use of Zoom also facilitated presentation of the stimulus material in the design workshop as well as recording quality.

However, we did experience some difficulties with participant engagement over Zoom. While we wanted to acknowledge the constraints of remote learning, it was challenging to facilitate conversations with participants who had their cameras off. We believe that it may be beneficial to consider requiring participants to turn on their cameras for future focus groups to increase overall engagement in group discussions. By more closely mirroring the affordances of face-to-face interaction, this may reduce some of the awkwardness of accidental interruptions and hesitation to speak, but it may also discourage personal disclosures. To address these challenges, we sought to build a rapport with participants by mirroring their words, referencing ideas that the group had mentioned earlier, and sharing personal anecdotes relating to the discussion. We found that, to a large extent, these strategies often encouraged participants to share their own experiences or thoughts on different topics.

6 CONCLUSION AND FUTURE WORK

Through four focus groups, the present study investigated how individuals and communities used social media to prepare for disasters, and how the design of these platforms influenced the formation and maintenance of social capital and disaster resilience among local communities. Our findings provide support for the role of social media in facilitating dialogic communication among community members, which in turn suggests the emergence of social media sites as both information sources and support networks for at-risk communities. The results of our paper inform the design of social media sites to support seamlessness across multiple platforms. Although we focus specifically on social media and disaster preparation, we expect that our findings may be applied to other contexts where individuals need to quickly search for information or communicate with others.

However, our study may be limited in that our sample is biased towards young people and students. We believe that our sample is representative of our target user group, as 90% of Americans aged 18 to 29 use at least one social media site [13], but our findings may lack generalizability for users who are less familiar with social media or live outside the United States. Future work should include more research and interviews with different demographics of users to gain a more detailed understanding of social media usage during disasters. While we did not probe deeply about how participants interacted with the design of different user interfaces in this study, we believe that future research should more closely examine how specific features and interactions within these interfaces (e.g., using color to convey information order in the FEMA app) influence disaster preparation on social media.

Additionally, future research may investigate the role of non-governmental agencies (NGOs) and other disaster relief

organizations in helping at-risk communities prepare for disasters, as well as how the design of social media platforms can enable these organizations to more effectively disseminate information to communities during disasters. Future work may also explore how digital volunteers can contribute to disaster preparedness efforts and whether social capital and community resilience function similarly or differently on social media when users are geographically dispersed.

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A APPENDICES

A.1 Characteristics of study participants

Table 1: Characteristics of study participants

Focus Group	Participant	Gender	Year	Major
1	1	Female	Sophomore	Pre-Communication
	2	Male	Junior	Pre-Communication
	3	Female	Sophomore	Pre-Communication
	4	Female	Junior	Communication
	5	Female	Junior	Film & Media Studies
	6	Female	Sophomore	Pre-Communication & Pre-Sociology
	7	Female	Freshman	Dance
2	8	Female	Senior	Communication
	9	Female	Freshman	Pre-Communication
	10	Male	Freshman	Pre-Economics
	11	Female	Graduate	Communication
3	12	Female	Freshman	Pre-Communication
	13	Female	Junior	Pre-Communication
	14	Male	Sophomore	Music & Pre-Communication
	15	Male	Junior	Pre-Communication
	16	Female	Junior	Psychology & Brain Sciences
	17	Female	Sophomore	Pre-Communication
	18	Male	Junior	Applied Math
	19	Female	Sophomore	Communication
4	20	Male	Sophomore	Linguistics & Communication
	21	Female	Freshman	Dance & Communication
	22	Female	Junior	Communication
	23	Female	Junior	Communication
	24	Female	Freshman	Undeclared
	25	Female	Sophomore	Economics & Accounting
	26	Female	Senior	Communication & Geography
	27	Female	Freshman	Film & Media Studies
	28	Female	Junior	Communication
	29	Male	Junior	Communication
	30	Female	Sophomore	Communication
	31	Male	Freshman	Undeclared